

SmartDocQA: A PDF-Centric Information Retrieval and Answer Extraction System

Abstract:

The rapid growth of digital documents has created a need for intelligent systems that can efficiently retrieve and extract relevant information from large collections of files. This project proposes SmartDocQA, a PDF-centric information retrieval and answer extraction system designed to automatically analyze PDF documents and provide precise answers to user queries. The system integrates Natural Language Processing (NLP), document parsing, and machine learning techniques to convert unstructured PDF content into searchable and structured data.

SmartDocQA processes uploaded PDF documents by extracting text, cleaning and preprocessing the data, and indexing the content for efficient retrieval. When a user submits a query, the system applies semantic search and question-answering techniques to identify the most relevant sections of the document and extract accurate answers. Advanced embedding methods and similarity matching are used to improve retrieval accuracy and contextual understanding.

The proposed system aims to reduce the time required to manually search through lengthy documents and enhance knowledge discovery in domains such as education, research, legal documentation, and enterprise knowledge management. Experimental results demonstrate that SmartDocQA can effectively retrieve relevant information and generate context-aware answers from PDF documents, improving user productivity and accessibility to large document repositories.

The system can be implemented using modern AI frameworks and integrated into web-based platforms, enabling users to interact with document collections through a simple and intelligent question-answering interface.

Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: At least 50 GB free space
- System Type: 64-bit Computer

Software Requirements

- Operating System: Windows 10 / Linux / macOS
- Programming Language: Python 3.8 or above
- IDE / Editor: VS Code / PyCharm / Jupyter Notebook
- Web Framework: Flask
- PDF Processing Library: PyPDF2 or pdfplumber
- Deep Learning / QA Models: Hugging Face Transformers